



MMA093 Rigid Body Dynamics

Assignment 3

Mandatory Part

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MMA093 Rigid Body Dynamics

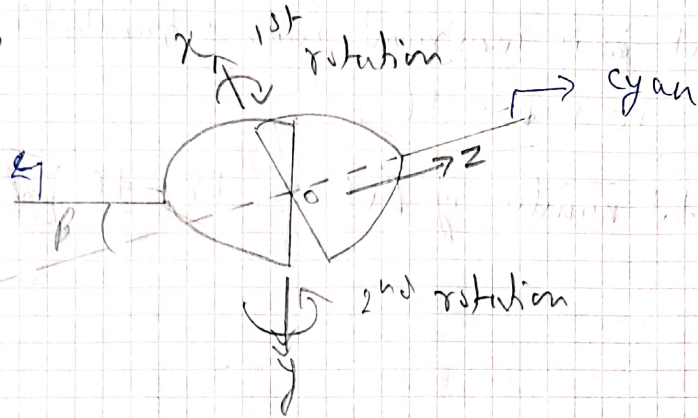
December 3, 2025

Problem 1:

Assignment 3

2.

Red $\rightarrow$



Consider a System, Consisting of 3 rigid bodies

Red is output shaft

Cyan is Input shaft.

Yellow is Hook joint.

Let 'o' be the point which is common to bodies output & input.

x-axis placed in the direction of 1st rotation axis.

y-axis placed in the direction of 2nd rotation axis

Let β_1 be the angle of rotation of body red around y-axis

Let β_2 be the angle of rotation of body cyan around x-axis.

$$6 \times 3 = 18 \text{ - DOF's.}$$

Input & output has revolute joint remove 5 DOF's,
 $2 \times 5 = 10$.

Hook joint is constraint with 3 translation & rotation
 $10 + 6 = 16$
 Total number of DOF's = $18 - 16 = \underline{\underline{2}}$.

Problem 2: Model of Universal Joint in ADAMS

The 3 Rigid bodies is considered in a system. Red shaft is considered as output shaft and Cyan shaft is considered as input shaft and yellow is a hook joint.

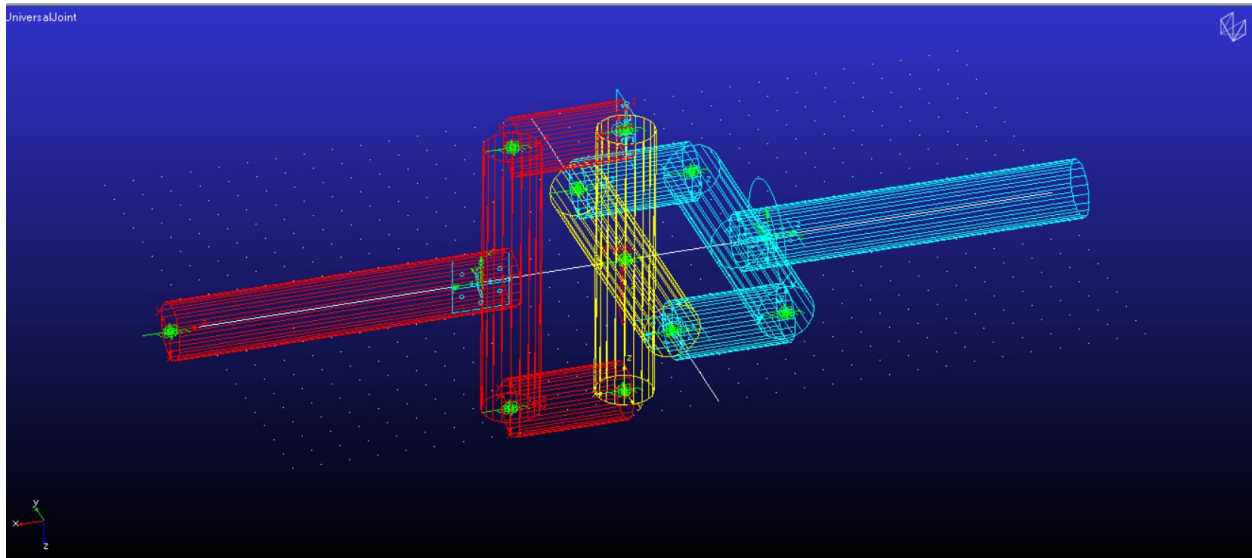


Figure 1: Model of Universal joint

The input angle is give to the input shaft based on which output angular speed, Speed ratio, analytical relation is compared for different Beta angle.

CASE 1: $\beta = 0\text{deg}$

The input and output angular speed is 30(deg/sec), Speed ration and analytical measure is 1.0

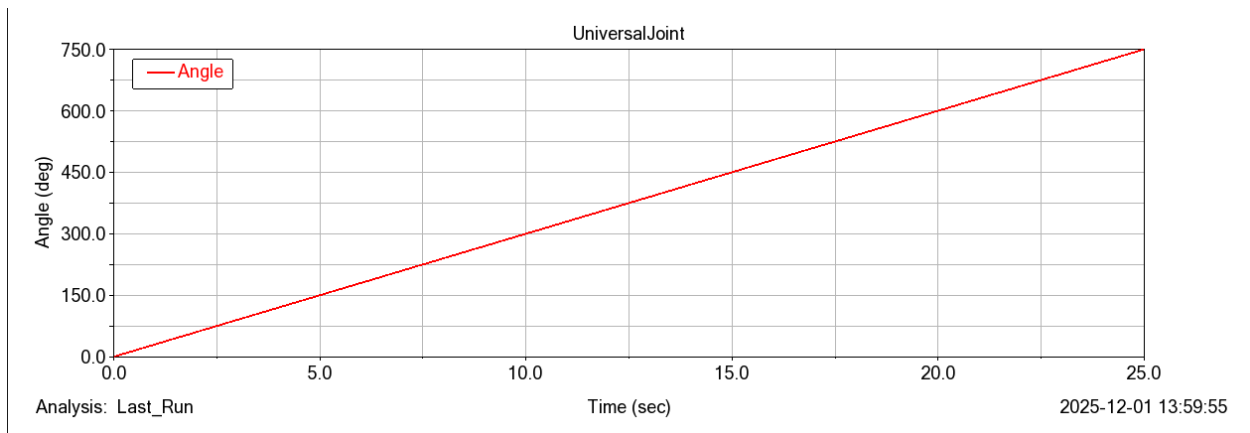


Figure 2: Input angle

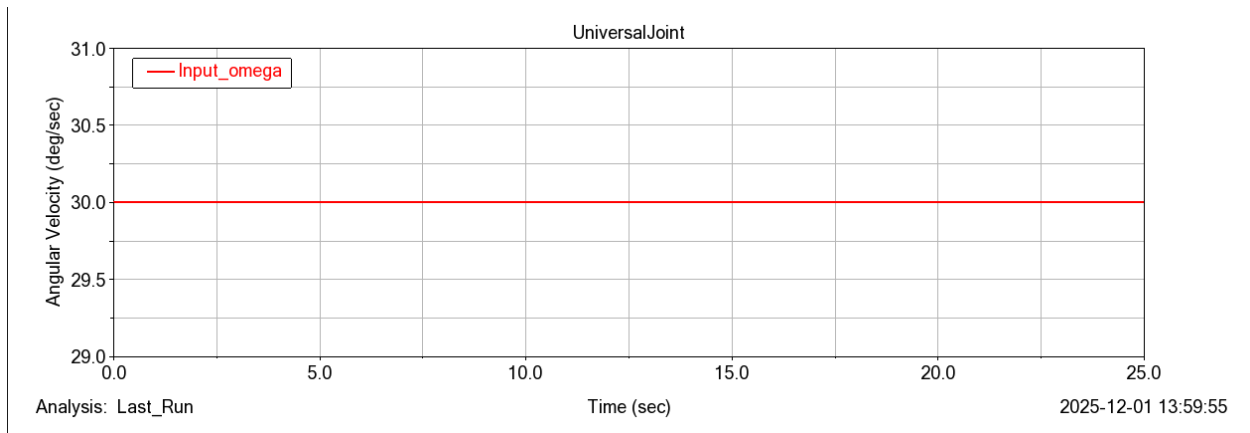


Figure 3: Input angular speed

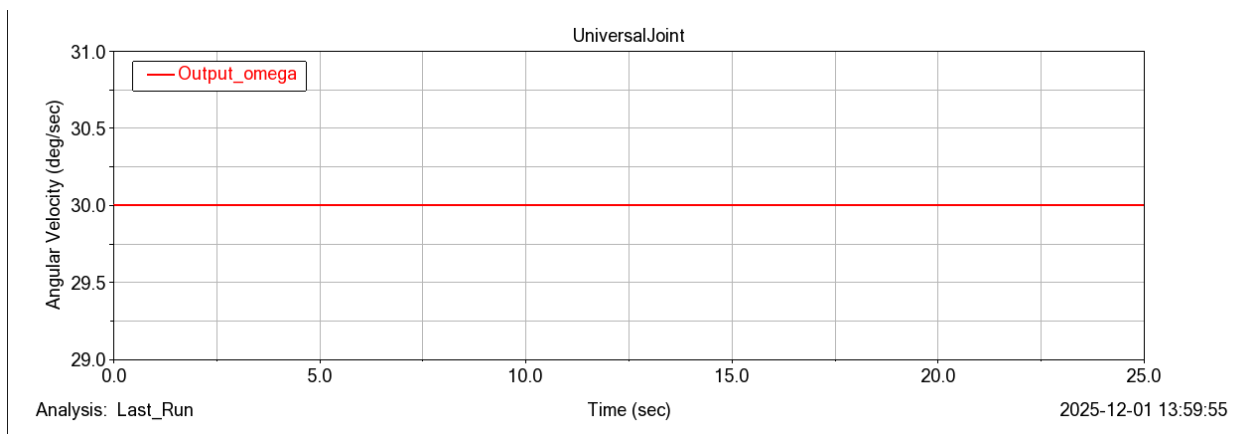


Figure 4: Output angular speed

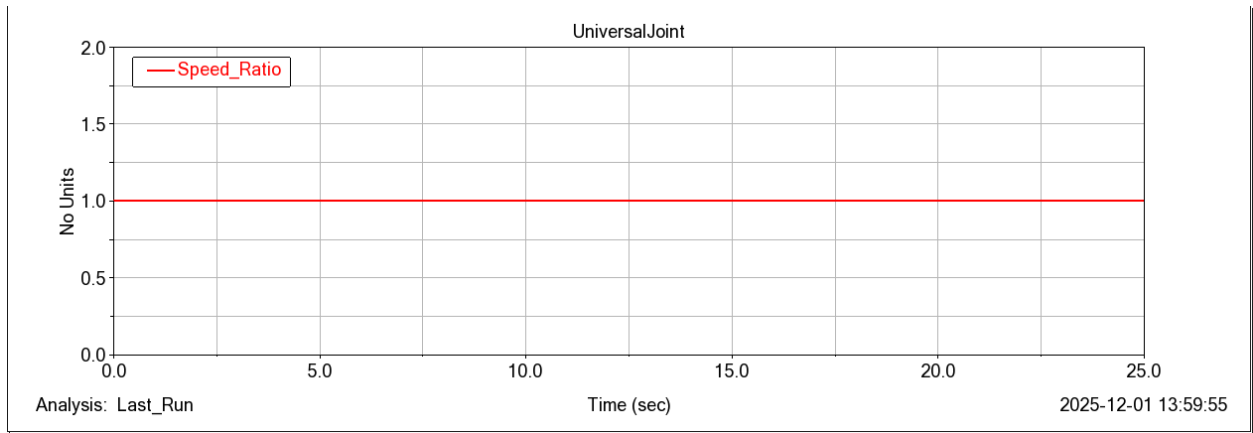


Figure 5: Speed Ratio

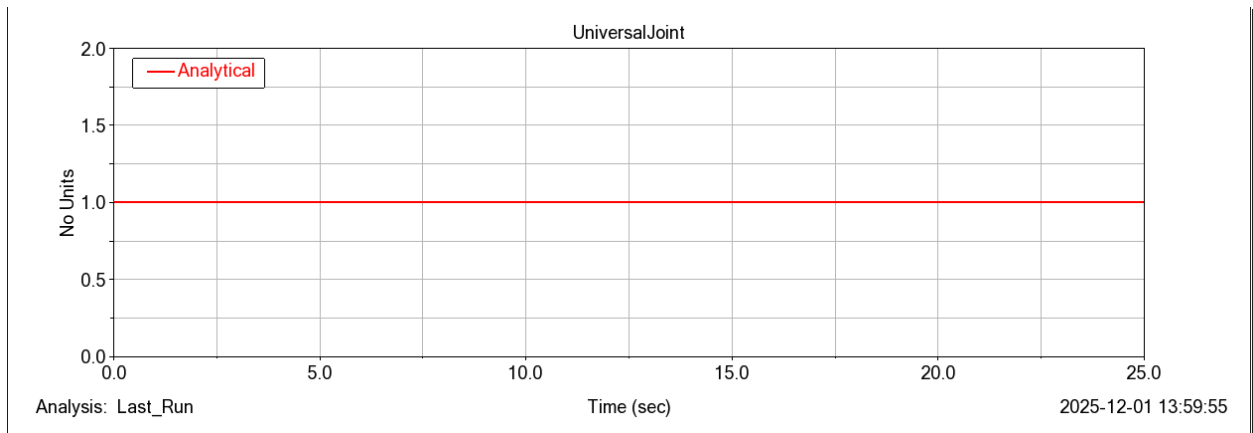


Figure 6: Analytical measure

CASE 2: $\beta = 10$ deg

The input angular velocity is 30(deg/sec) and output angular speed varies between max and min 30.46(deg/sec) to 29.54(deg/sec), Speed ration and analytical measure varies between 1.015 to 0.984.

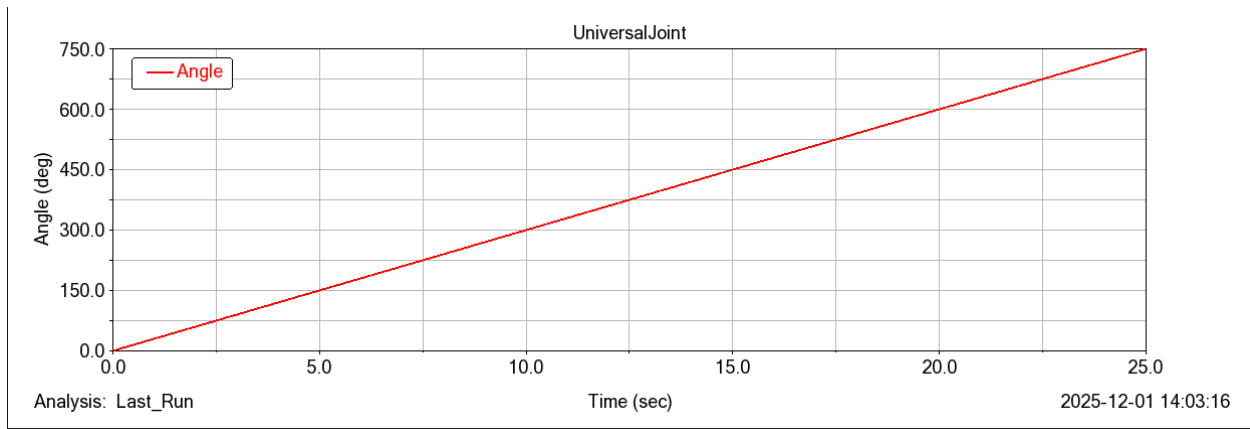


Figure 7: Input angle

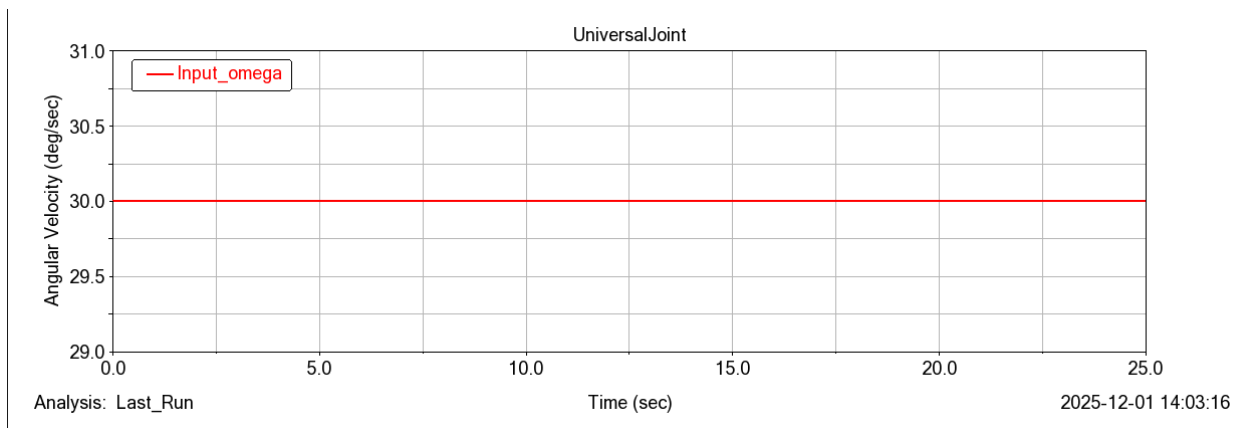


Figure 8: Input angular speed

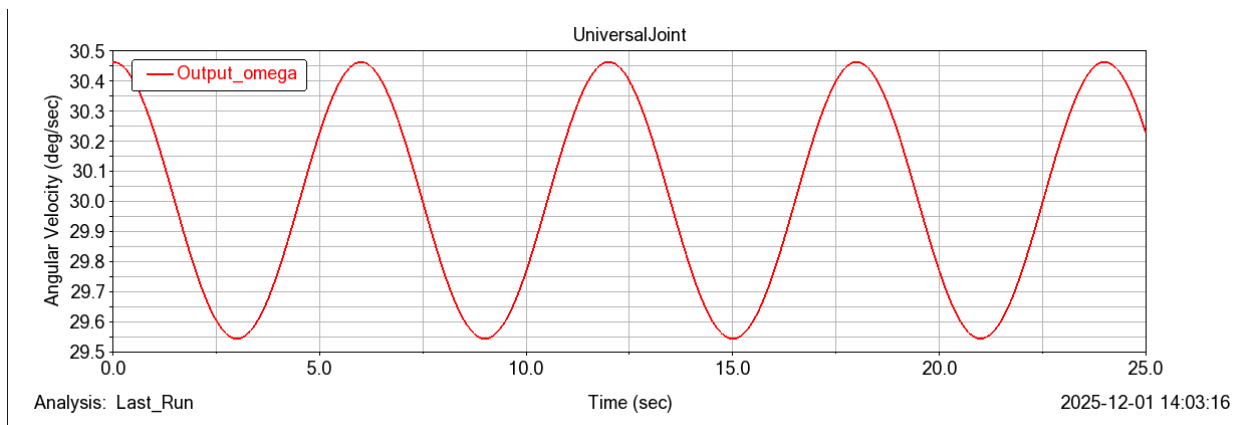


Figure 9: Output angular speed

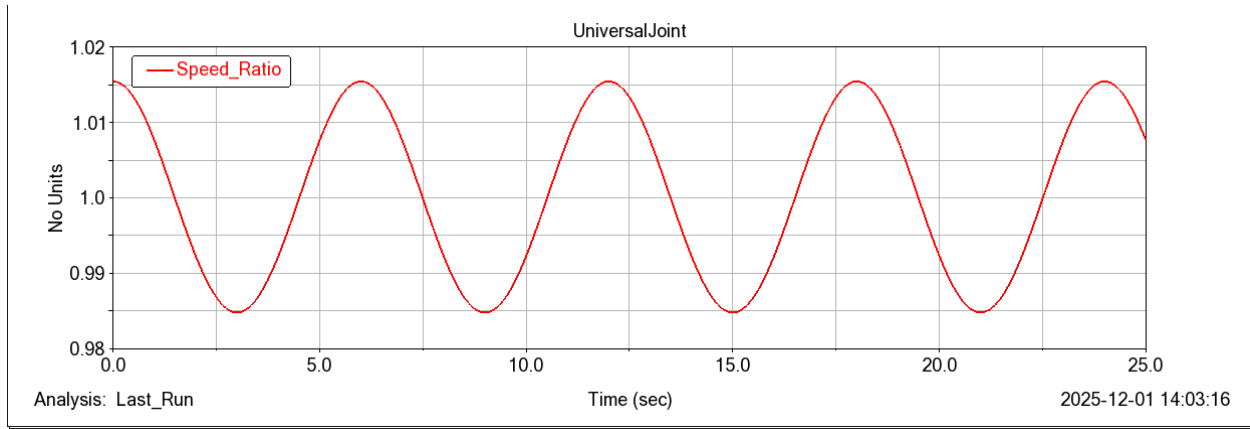


Figure 10: Speed Ratio

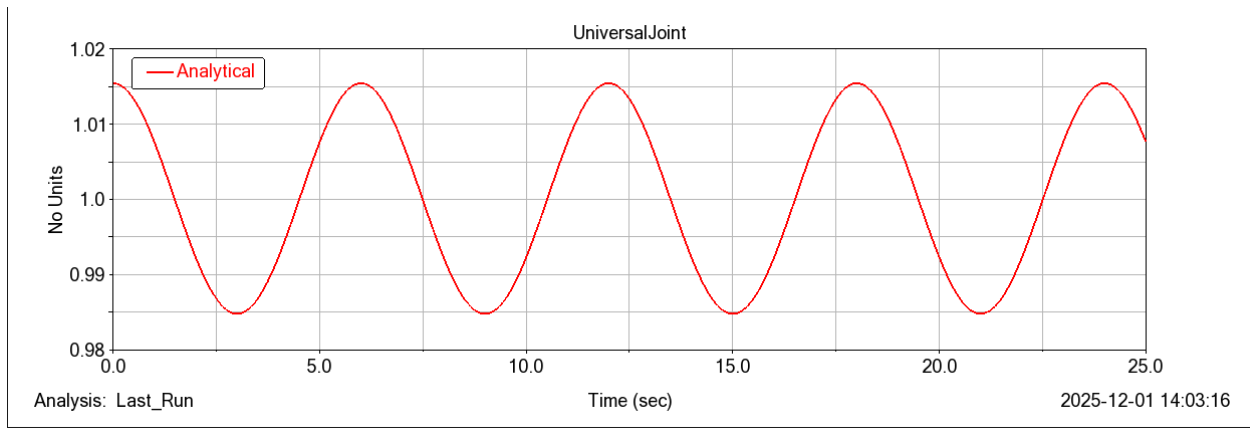


Figure 11: Analytical measure

CASE 3: $\beta = 30$ deg

The input angular velocity is 30(deg/sec) and output angular speed varies between max and min 34.64(deg/sec) to 29.98(deg/sec), Speed ration and analytical measure varies between 1.154 to 0.866.

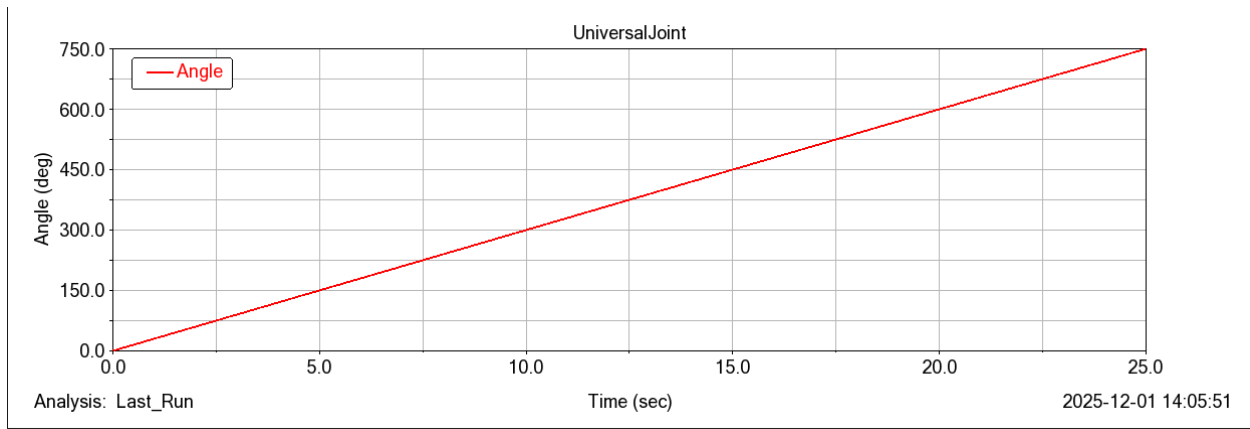


Figure 12: Input angle

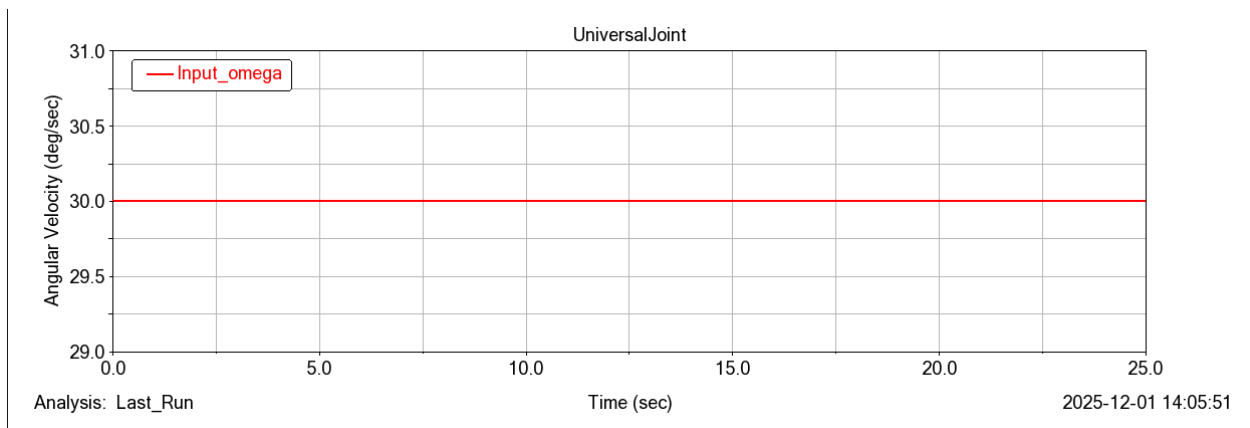


Figure 13: Input angular speed

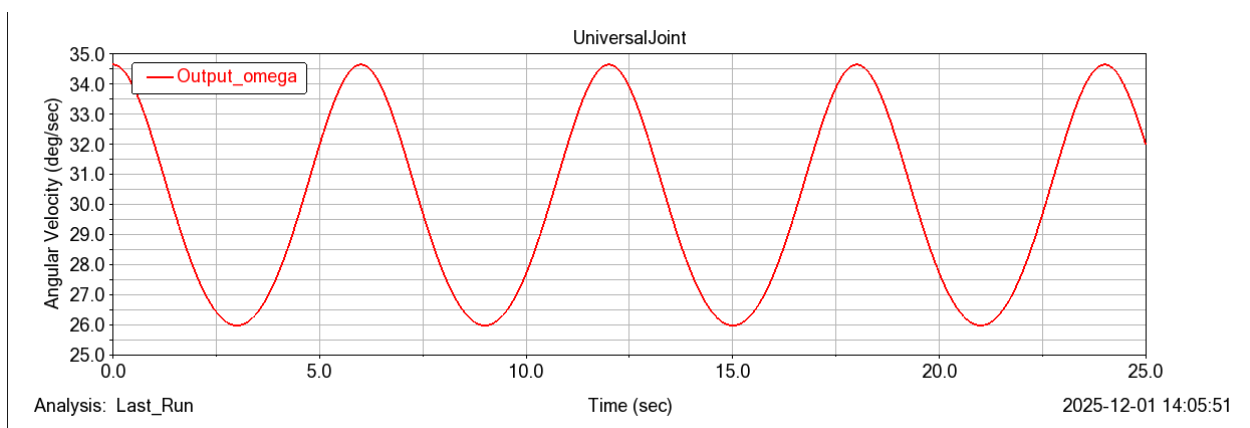


Figure 14: Output angular speed

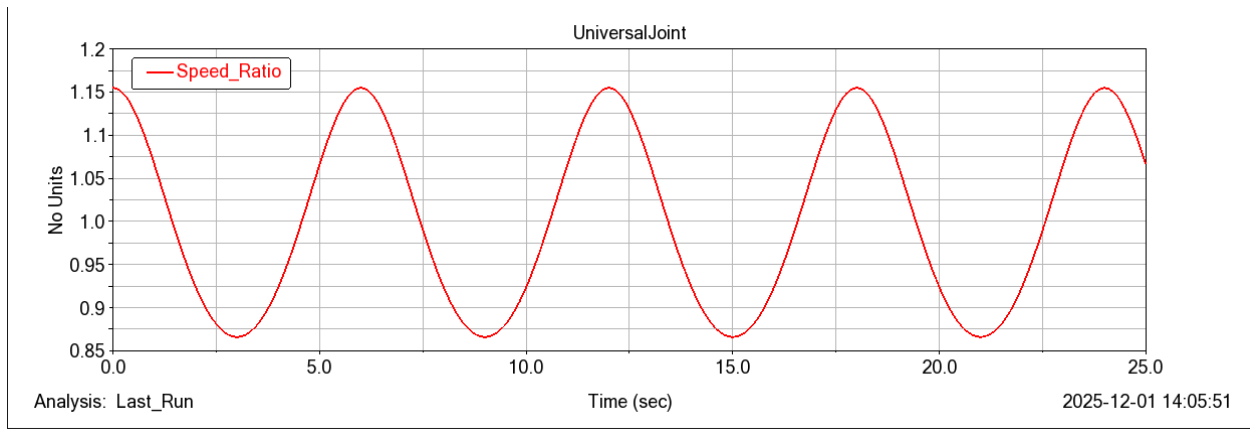


Figure 15: Speed Ratio

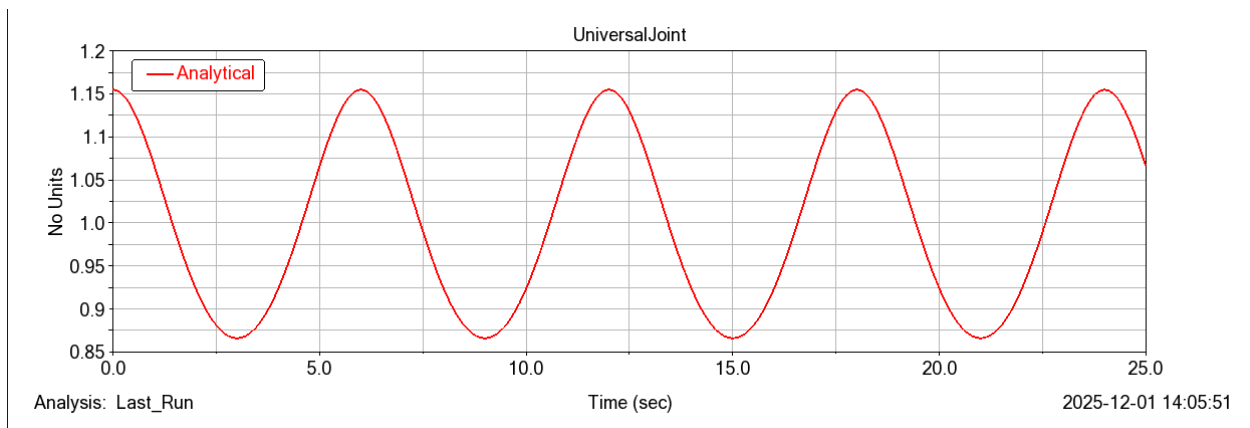


Figure 16: Analytical measure

Problem 3. ADAMS Built in hook Joint

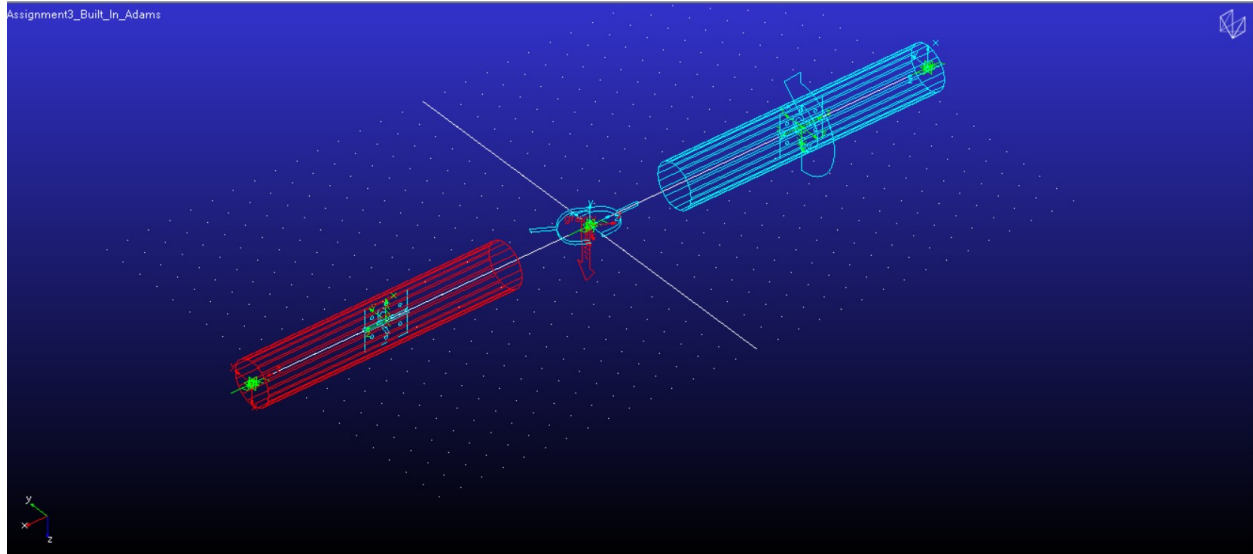


Figure 17: Built in Hook joint

The Red shaft is Output shaft and cyan shaft is Input shaft. In this problem the Input and output shaft is connected together by a Built in hook joint in ADAMS. The values for Output angular speed, speed ratio and analytical is measured for all 3 cases.

CASE 1: $\beta = 0\text{deg}$

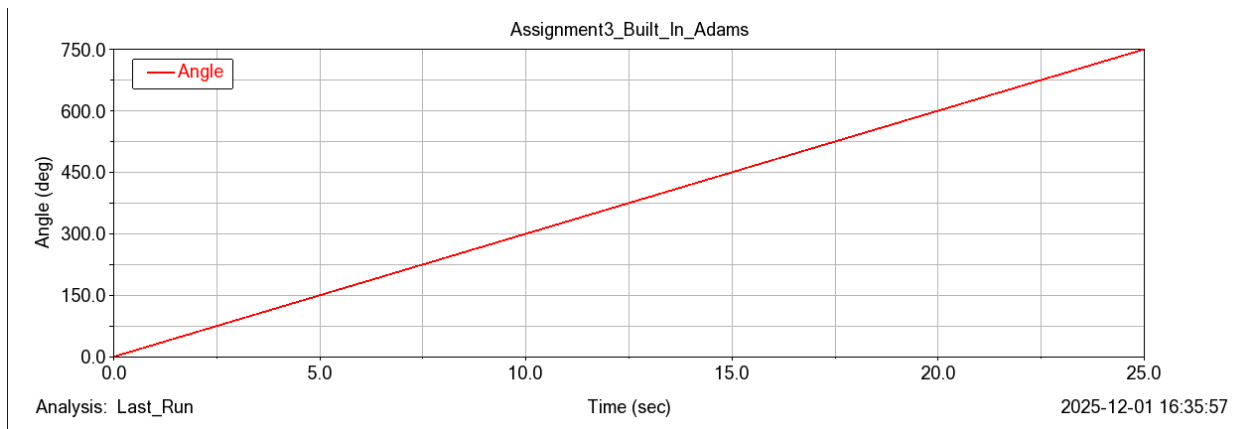


Figure 18: Input angle

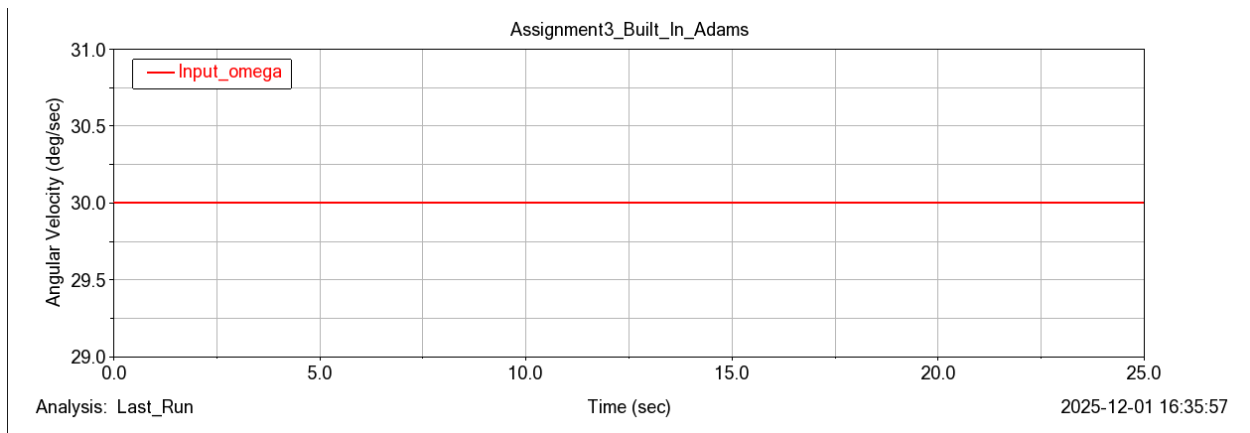


Figure 19: Input angular speed

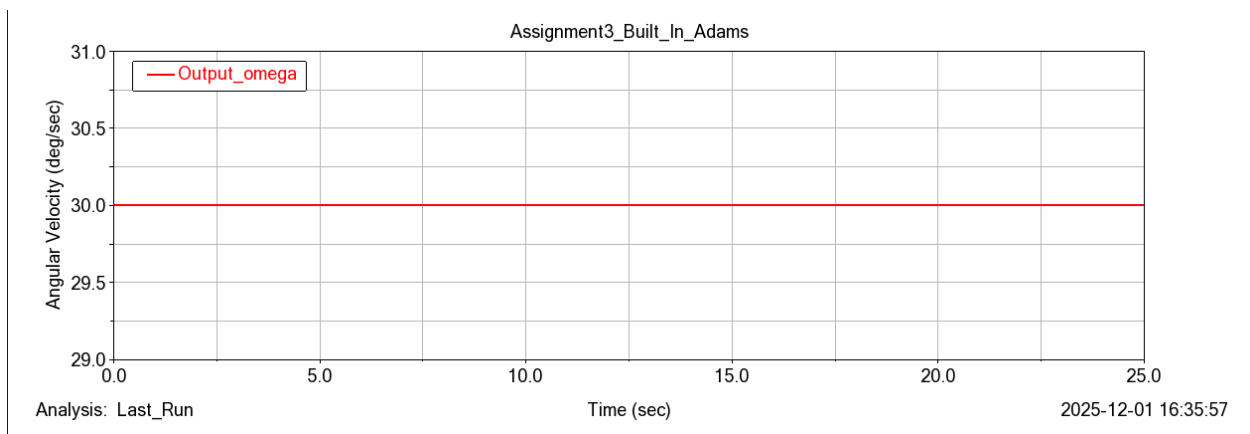


Figure 20: Output angular speed

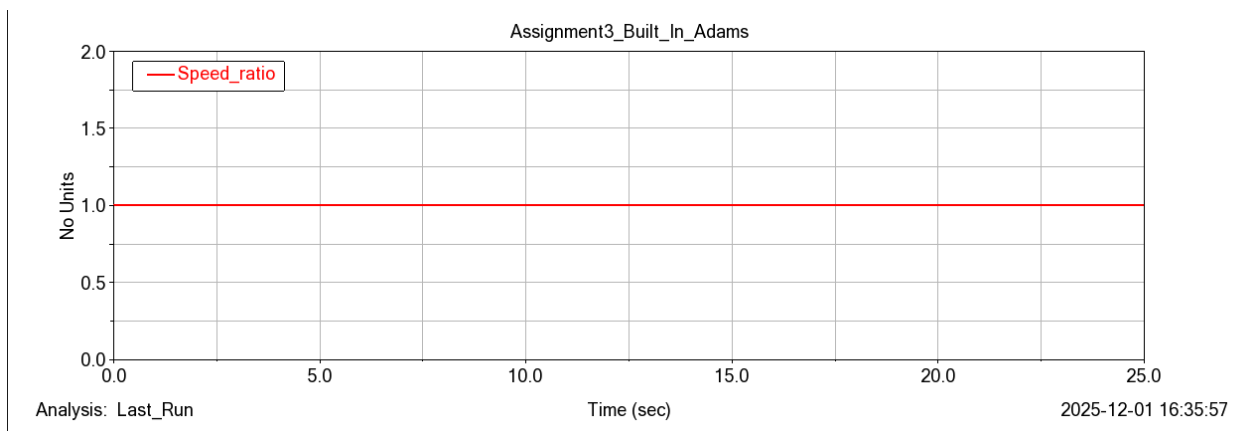


Figure 21: Speed Ratio

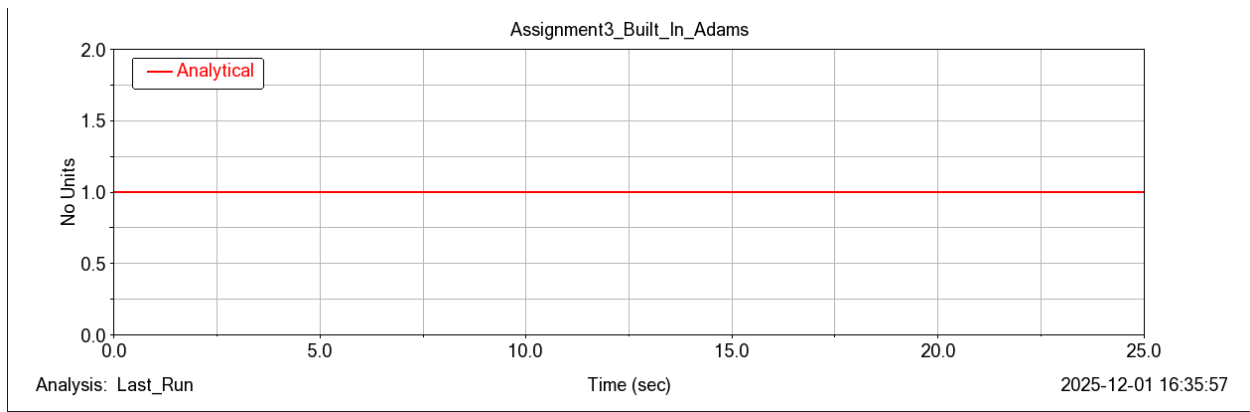


Figure 22: Analytical measure

CASE 2: $\beta = 10$ deg

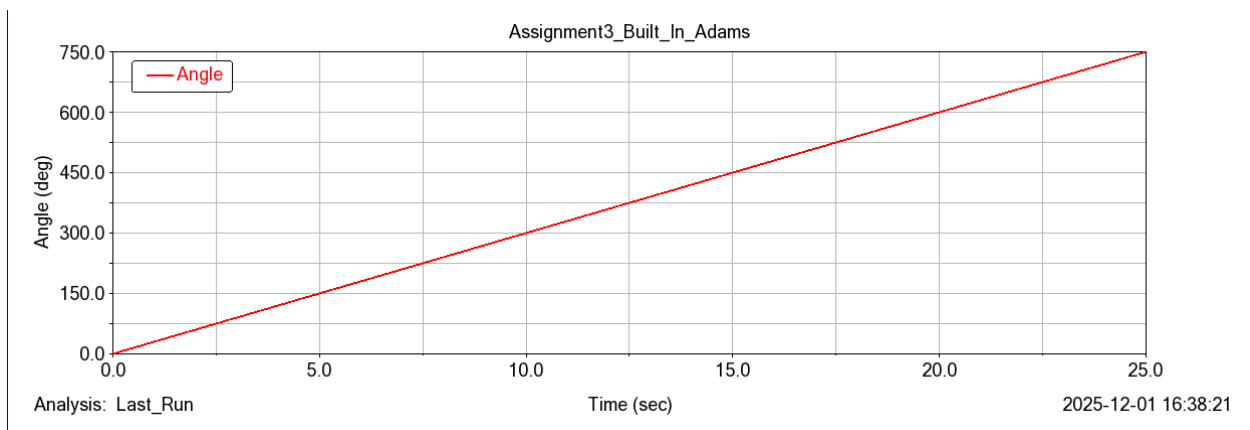


Figure 23: Input angle

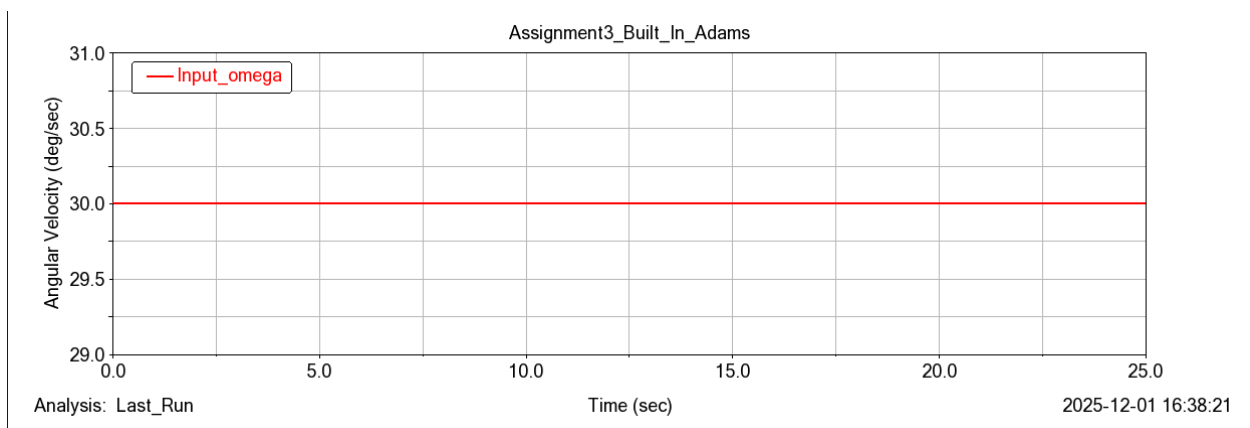


Figure 24: Input angular speed

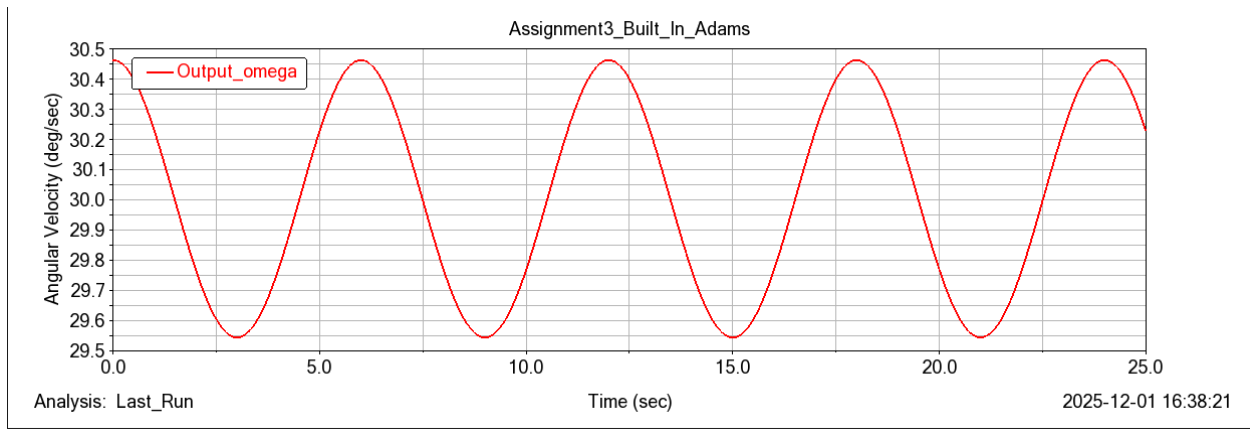


Figure 25: Output angular speed

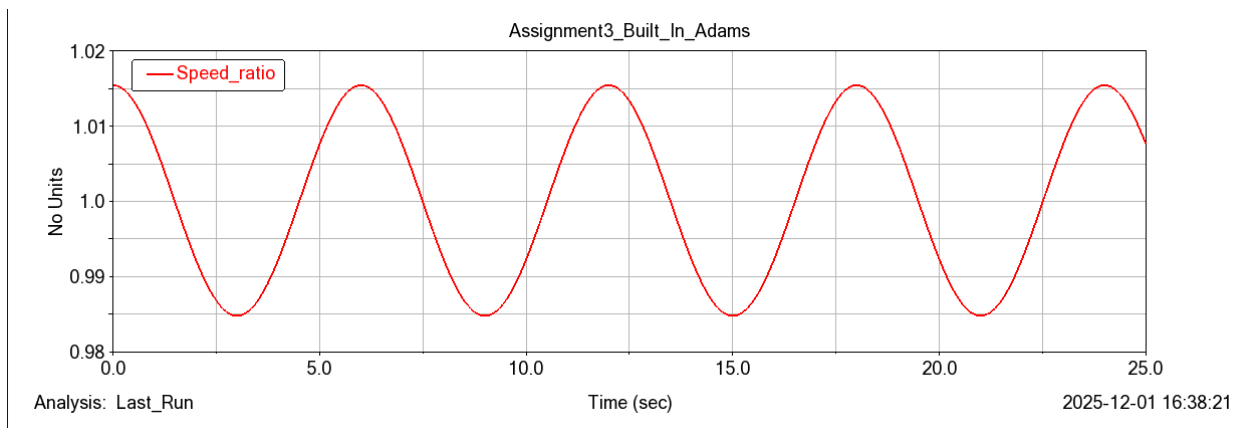


Figure 26: Speed Ratio

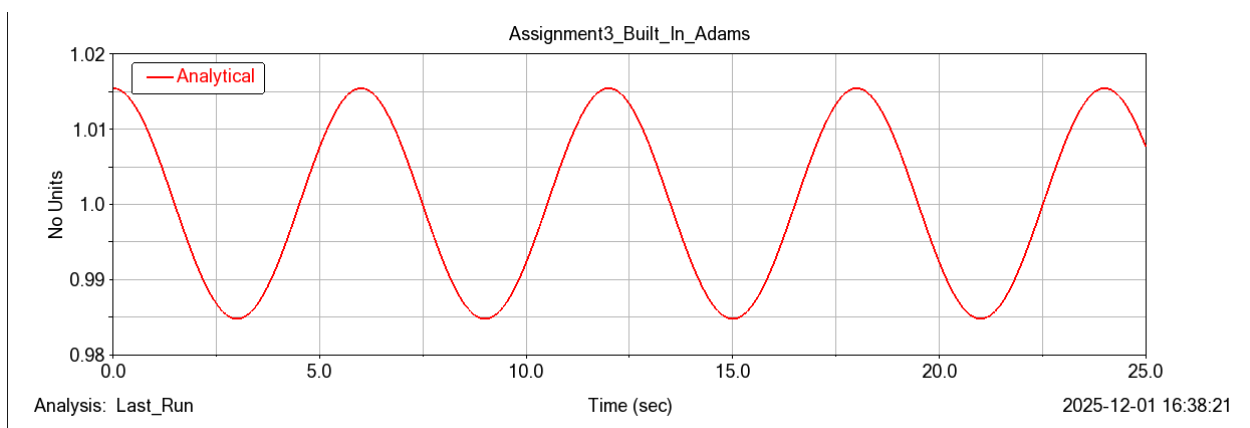


Figure 27: Analytical measure

CASE 3: $\beta = 30$ deg

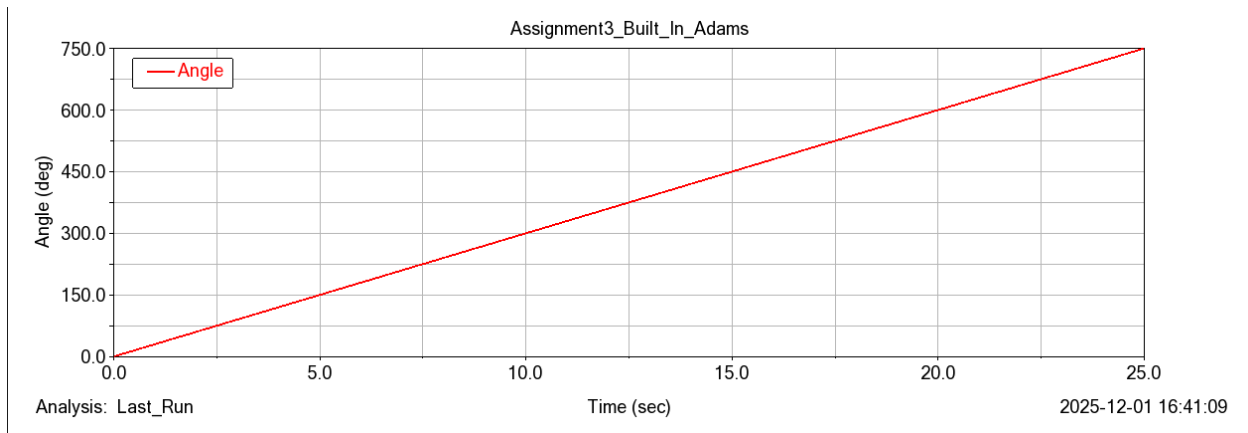


Figure 28: Input angle

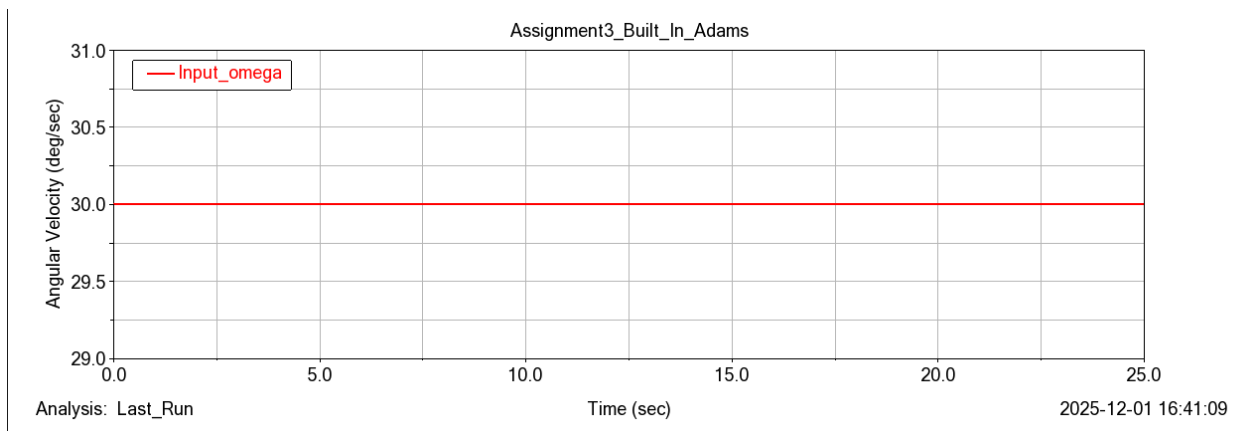


Figure 29: Input angular speed

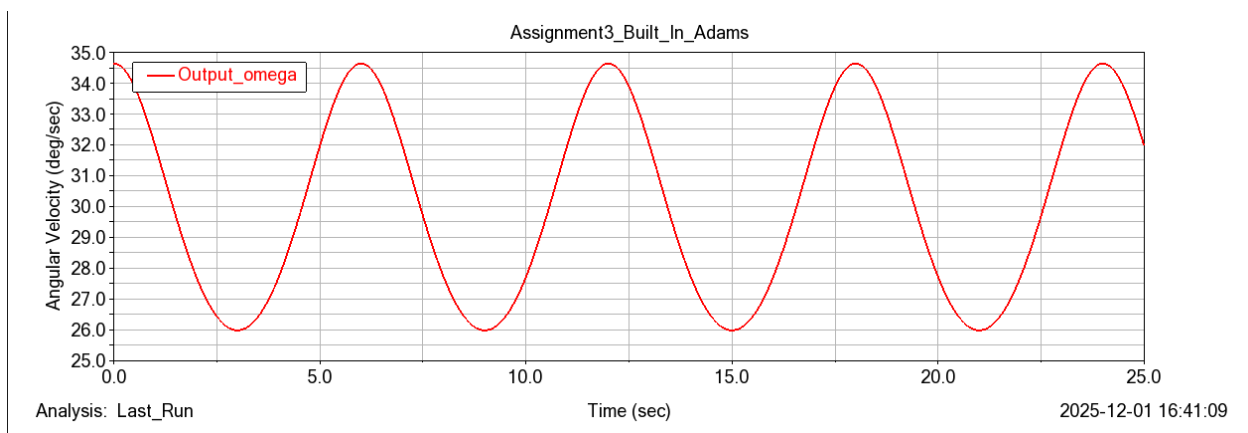


Figure 30: Output angular speed

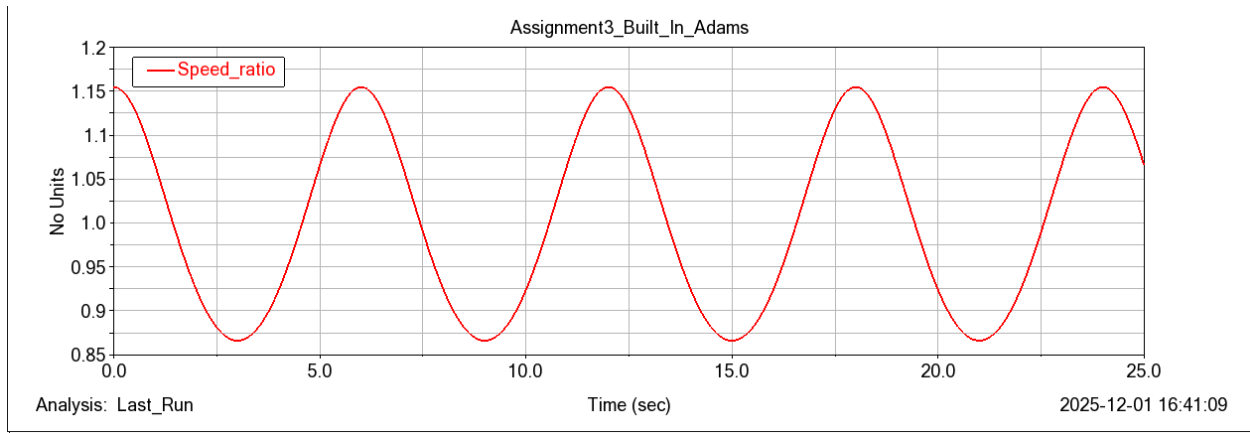


Figure 31: Speed Ratio

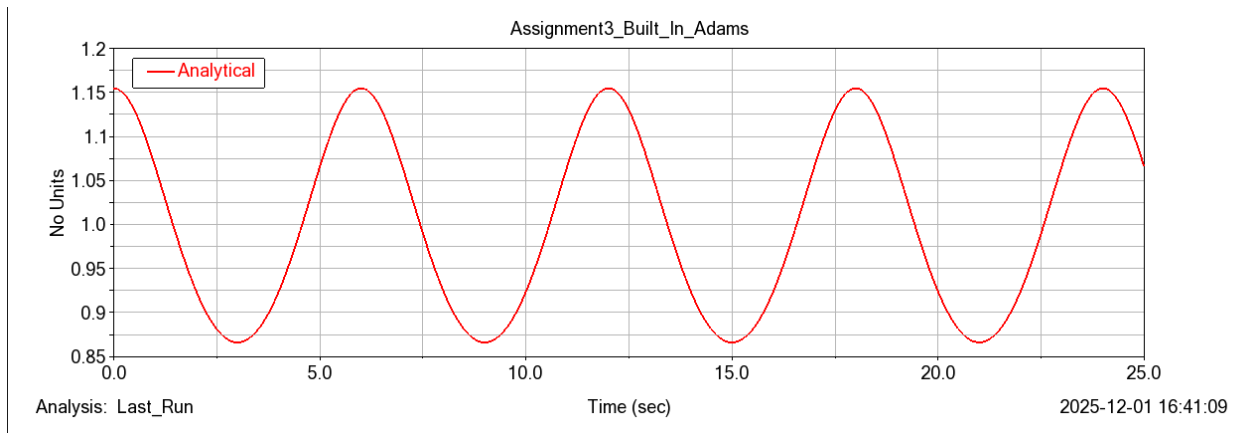


Figure 32: Analytical measure

In all the 3 cases $\beta = 0, 10, 30$ deg, the input angular speed, output angular speed, speed ratio, analytical measure values is similar to the problem 2.